

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-10 09:30 UTC

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Generated 2026-05-10 09:30 UTC

- Cycles recorded: **534**
- Time span: 395.8h (~1.35 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	499
BARRIER_HIT	16
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	29
Free Probability	12
Schur-Weyl Duality	12
Representation Theory of Symmetric Groups	11
Matroid Theory	10
Algebraic Geometry	7
Additive Combinatorics	7
Polymatroid Theory	7
Random Matrix Theory	6
Finite Geometry	6
Spectral Graph Theory	6
Plethysm Theory	6
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4
Algebraic Geometry of Secant Varieties	4
FOURIER_ANALYSIS	4
Fourier analysis of boolean functions	3
Algebraic Topology	3

Field	Cycles
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3
Schur-Weyl duality	3
Noncommutative L^p Geometry	3
Persistent Homology	3
Noncommutative Geometry	3
Noncommutative geometry	2
Representation theory of symmetric groups	2
Ergodic Circuit Framework (COMM_COMPLEXITY)	2

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-10 03:58 UTC	INCONCLUSIVE	Hypergraph Maximum Matching Inverse Proportional to ACC^0 Circuit
2026-05-10 04:08 UTC	INCONCLUSIVE	Finite Field Rank and Branching Program Width for Boolean Functio
2026-05-10 04:25 UTC	INCONCLUSIVE	Matroid Rank Inverse Proportional to ACC^0 Circuit Size for $GF(2)$
2026-05-10 04:55 UTC	INCONCLUSIVE	Secant Variety Dimension Lower Bounds Disjointness Communication
2026-05-10 05:06 UTC	INCONCLUSIVE	Real Critical Point Count Exponential in SOS Degree for Max-CUT
2026-05-10 05:57 UTC	INCONCLUSIVE	Schatten p -Norm Lower Bound for Disjointness
2026-05-10 06:28 UTC	INCONCLUSIVE	Communication Comple SOS Degree Lower Bound via Adjacency Eigenvalue Multiplicity
2026-05-10 07:07 UTC	INCONCLUSIVE	Additive Energy Lower Bound via Read-Twice Branching Program Disc
2026-05-10 07:15 UTC	INCONCLUSIVE	Real Root Count Exponential in AC^0 PARITY Circuits
2026-05-10 07:36 UTC	INCONCLUSIVE	Polymatroid Rank Lower Bound for Monotone CLIQUE
2026-05-10 07:48 UTC	INCONCLUSIVE	Modular Character Degree Gap in Permanent vs Determinant Circuits

Time	Verdict	Title
2026-05-10 08:11 UTC	INCONCLUSIVE	Class Number Inverse Proportional to Resolution Length in 3-SAT
2026-05-10 08:23 UTC	INCONCLUSIVE	BIBD Incidence Matrix ACC ⁰
2026-05-10 09:07 UTC	INCONCLUSIVE	Circuit Size Lower Bound Minimal Incidence Count Inverse Proportional to Disjointness Comm
2026-05-10 09:21 UTC	INCONCLUSIVE	Standard Young Tableau Count Exponential Gap in Symmetric Power D

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
  pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```